



MSDS Report

This SDS is prepared in accordance with OSHA HazCom 2012 (29 CFR 1910.1200)

Client	SHENZHEN FUERXIN TECHNOLOGY CO.,LTD.
Add. of Client	5/F, 1st Building, NO.264, Baoshi East Road, Shuitian Community, Shiyan Town, Baoan District, Shenzhen Guangdong, P.R.China
Sample Name	Li-ion Battery
Sample Model	FEX 2560110 20000mAh
Nominal voltage	3.7V
Rated capacity	20000mAh, 74Wh
Sample Mass	327.8g
Sample Size	(114.0×61.0×26.0)mm(L×W×T)
Prepared by	Nuocheng Safety Testing (Shenzhen) Co., Ltd. 1-2/F., No.A Building, The Third Industrial Zone, Fenghuang Community, Fuyong Street, Baoan District, Shenzhen, Guangdong, China
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Seal of



Material Safety Data Sheet

Section 1 - Product & Company Identification

Sample Name	Li-ion Battery
Manufacturer	SHENZHEN FUERXIN TECHNOLOGY CO.,LTD.
Address	5/F, 1st Building, NO.264, Baoshi East Road, Shuitian Community, Shiyan Town, Baoan District, Shenzhen Guangdong, P.R.China
Factory	SHENZHEN FUERXIN TECHNOLOGY CO.,LTD.
Address	5/F, 1st Building, NO.264, Baoshi East Road, Shuitian Community, Shiyan Town, Baoan District, Shenzhen Guangdong, P.R.China
Emergency Tel.	+86-755-29452478
North America 24-Hour Chemtrec Emergency Hotline	1-800-424-9300
Email address	Fuerxin6@126.com

Section 2 - Hazard Identification

GHS Hazard Classification	Skin Corrosion / Irritation Category 2 Serious Eye Damage Category 1 Specific Target Organ Toxicity Category 3 Hazardous to the Aquatic Environment, Chronic Category 4
Signal Word	Danger (Only for damaged leaking cells; intact sealed battery presents no hazard under normal use)
Hazard Statements (H-Codes)	H315: Causes skin irritation H318: Causes serious eye damage H335: May cause respiratory irritation H413: May cause long lasting harmful effects to aquatic life
Precautionary Statements (P-Codes)	Prevention P264: Wash skin thoroughly after handling leaked material P280: Wear protective gloves/eye protection/face protection P271: Use only outdoors or in a well-ventilated area P273: Avoid release to the environment P222: Keep away from air if casing is broken P210: Keep away from heat, hot surfaces, sparks, open flames Response P302+P352: IF ON SKIN: Wash with plenty of water P305+P351+P338: IF IN EYES: Rinse continuously with water for several minutes; remove contact lenses if present P304+P340: IF INHALED: Remove person to fresh air and keep comfortable for breathing P310: Immediately call a poison center/doctor P391: Collect spillage of leaked electrolyte Storage

	P403+P233: Store in a well-ventilated place; keep tightly sealed P410+P403: Protect from sunlight and store in a well ventilated area below 60°C Disposal P501: Dispose of battery/leaked residue in accordance with local, state
Hazard Pictograms:	Corrosion, Exclamation Mark, Environment
Signal Word:	DANGER

Section 3 - Composition/Information on Ingredients

Chemical Name	Concentration or concentration ranges (%)	CAS Number/CAS号
Lithium Cobalt Oxide	35-38	12190-79-3
Graphite	20-22	7782-42-5
Copper	9-10	7440-50-8
Aluminum	5-6	7429-90-5
Ethylene carbonate	14-16	96-49-1
Polypropylene	5-6	9003-07-0
Carbonate, methyl ethyl	4-5	623-53-0
Phosphate(1-), hexafluoro-, lithium	5-6	21324-40-3

Section 4 - First Aid Measures

Inhalation	Remove source of contamination or move victim to fresh air. Obtain medical advice.
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Ingestion	Please rinse mouth thoroughly with water. Induce vomiting under the guidance of professional personage. Please seek medical treatment in time.
Skin contact	Take off contaminated clothing, rinse with plenty of water, if skin allergies and redness, please seek medical treatment in time.
Eye contact	Irrigate with flowing water for 15 minutes. If irritation persists, please seek medical treatment in time.

Section 5 - Fire Fighting Measures

Characteristics of Hazard	Toxic fumes, gases or vapors may evolve on burning.
Hazardous Combustion Products	Carbon monoxide, carbon dioxide, oxide fumes and so on.
Fire-extinguishing Methods and Extinguishing Media	For small fires, can use water, dry sand and other proper fire extinguishing media.
Attention in Fire-extinguishing	Firefighting personnel shall wear gas masks and full-body firefighting suits.

Section 6 - Accidental Release Measures

Personal Precautions, protective equipment, and emergency procedures	In case of rupture. Attention! Corrosive material. Avoid contact with skin, eyes and clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Refer to protective measures listed in Sections 7 and 8.
Environmental Precautions	Prevent product from contaminating soil and from entering sewers or waterways.

Methods and materials for Containment	Stop the leak if safe to do so. Contain the spilled liquid with dry sand or earth. Clean up spills immediately.
Methods and materials for cleaning up	Absorb spilled material with an inert absorbent (dry sand or earth). Scoop contaminated absorbent into an acceptable waste container. Collect all contaminated absorbent and dispose of according to directions in Section 13. Scrub the area with detergent and water; collect all contaminated wash water for proper disposal.

Section 7 - Handling and Storage

Handling	Do not disassemble, squeeze, or put the battery into a high temperature environment. Do not short circuit or install with incorrect polarity.
Storage	Store in a cool, dry, well-ventilated area away from incompatible substances. Keep out of the reach of children.
Other Precautions	Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Use personal protection equipment.

Section 8 - Exposure Controls/Personal Protection

Engineering Controls	Use local exhaust ventilation if battery cells rupture and electrolyte leaks during processing. Prevent accumulation of airborne metal dust.
Exposure Limits: Cobalt and its compounds:	ACGIH TLV: 0.02 mg/m ³ (inhalable fraction, TWA) OSHA PEL: 0.1 mg/m ³ TWA
Copper dusts and mists:	ACGIH TLV: 1 mg/m ³ TWA OSHA PEL: 1 mg/m ³ TWA

Aluminum:	ACGIH TLV: 10 mg/m ³ (total dust) TWA
Personal Protective Equipment	Respiratory protection: N95 respirator if high concentration dust/aerosol occurs under failure conditions. Hand protection: Nitrile resistant gloves. Eye protection: Safety goggles with face shield when handling damaged batteries. Skin protection: Impermeable work clothing.

Section 9 - Physical and Chemical Properties

Physical State	Form: Solid
	Color: Blue aluminum plastic composite film outer package
	Odour: Odorless under intact normal state; pungent acidic vapor if ruptured & electrolyte leaks
Change in condition	
pH	4.0-5.0
Flash Point	24 °C
Flammability	Non-flammable as sealed finished cell; electrolyte liquid flammable once package broken
Relative density	1.05 g/cm ³
Solubility (water)	Insoluble (Cell internal materials do not dissolve in water)

Section 10 - Stability and Reactivity

Chemical Stability	Stable under recommended storage conditions.
Conditions to Avoid	Heat above 70°C or incinerate, expose over a long period to humid conditions.
Incompatible materials	Acids, Oxidizing agents, Bases.
Hazardous Decomposition Products	Toxic Fumes, and may form peroxides.

Section 11 - Toxicological Information

Irritation	Vapors from ruptured battery cause eye & respiratory irritation; leaked electrolyte leads to severe skin corrosion.
Sensitization	No confirmed skin/respiratory sensitization under normal intact battery use.
Reproductive Toxicity	No reproductive hazard exposure for sealed, undamaged cells.
Toxicologically Synergistic Materials	No synergistic toxic effects observed for intact battery packs.

Section 12 - Ecological Information

General note	Prevent battery waste from entering groundwater, rivers and sewer systems.
Ecotoxicity:	Cobalt, copper and nickel heavy metals inside the cell are toxic to aquatic organisms if leaked into water environment; carbonate solvents cause chronic aquatic harm.

Persistence & Bioaccumulation:	Metal components do not biodegrade and accumulate in aquatic food chains.
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Section 13 - Disposal Considerations

Waste Treatment	Recycle or dispose of in accordance with government, state & local regulations.
Attention for Waste Treatment	Deserted batteries couldn't be treated as ordinary trash. Couldn't be thrown into fire or placed in high temperature. Couldn't be dissected, pierced, crushed or treated similarly. Best way is recycling.

Section 14 - Transport Information

UN number	3480 or 3481
Proper shipping name	Lithium ion batteries (including lithium ion polymer batteries) or; Lithium ion batteries packed with equipment (including lithium ion polymer batteries) or; Lithium ion batteries contained in equipment (including lithium ion polymer batteries).
DGR	Can be shipped by air in accordance with International Civil Aviation Organization (ICAO) or International Air Transport Association (IATA) DGR 67 Edition (2026) Packing Instructions Section IB of 965, Section II of 966 or Section II of 967 appropriately. For PI965 IB: Limited to a maximum of 30% SoC, and paste Class 9-Batteries Label, Cargo Aircraft Only Label and Battery Mark.

	For PI966 II: More than 2.7Wh, limited to a maximum of 30% SoC. Paste Battery Mark. For PI967 II: Paste Battery Mark. UN Classification (Transport hazard class): Class 9 (PI965 Section IB) or N/A (PI966~967 Section II) PG Packing Group: N/A
IMDG CODE	The batteries are not restricted to IMDG Code 2024 Edition (Amdt 42-24) according to special provision 188. Paste Battery Mark. UN Classification (Transport hazard class): N/A PG Packing Group: N/A Marine pollutant (Y/N): N EmS No.: F-A, S-I
US Domestic Ground Transport (49 CFR 173.185)	For domestic transportation within the United States under U.S. DOT Hazardous Materials Regulations 49 CFR Part 173.185: This FEX 2560110 lithium-ion polymer pouch cell (3.7V, 16000mAh, 59.2Wh) has passed UN 38.3 full testing, satisfying the qualification requirements of 49 CFR §173.185(a). Equipped with built-in multi-layer protection circuit (overcharge, over-discharge, short-circuit, over-temperature protection) and safety pressure relief valve, complying with 49 CFR §173.185(d) design standards for consumer lithium-ion batteries.

	When shipped contained inside portable cooling fan equipment, the product falls under simplified transport exemption provisions of 49 CFR 173.185 Special Provision 188 equivalent, no full Class 9 hazardous goods placard required for retail finished packages. Applicable UN numbers: UN 3480 (loose standalone battery cells) / UN 3481 (batteries contained in finished equipment). Hazard classification: Class 9 Miscellaneous Dangerous Goods; Packing Group: Not Applicable for factory-sealed consumer battery packs under domestic retail exemptions. Packaging requirements: All outer packaging shall prevent cell short-circuit, physical extrusion and high-temperature exposure, consistent with 49 CFR packaging performance criteria for lithium ion batteries.
ADR	The batteries are not restricted according to ADR 205 special provision 188. Paste Battery Mark. UN Classification (Transport hazard class): N/A PG Packing Group: N/A
RID	The batteries are not restricted according to RID 205 special provision 188. Paste Battery Mark. UN Classification (Transport hazard class): N/A PG Packing Group: N/A

Section 15 - Regulatory Information

Recommendations on the Transport of Dangerous Goods-Model Regulations

Recommendations on the Transport of Dangerous Goods-Manual of Tests and Criteria

Dangerous Goods Regulations

International Maritime Dangerous Goods Code

Regulations Concerning Road Transportation of Dangerous Goods

Agreement Concerning the International Carriage of Dangerous Goods by Road (ADR)

Regulations concerning the International Carriage of Dangerous Goods by Rail (RID)

Technical Instructions for the Safe Transport of Dangerous Goods

Classification and Code of Dangerous Goods

Occupational Safety and Health Act (OSHA)

Toxic Substance Control Act (TSCA)

Federal Environmental Pollution Control Act (FEPCA)

Resource Conservation and Recovery Act (RCRA)

Code of Federal Regulations (CFR)

In accordance with all Federal, State and local laws.

Section 16 - Other Information

To the best of our knowledge, the information contained herein is accurate. However, neither the above named supplier nor any of its subsidiaries assumes any liability whatsoever for the accuracy or completeness of the information contained herein. Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that these are the only hazards that exist.

The data/information contained herein has been reviewed and approved for general release on the basis that this document contains no export controlled information.

*******End of Report*******